

SECD2613

System Analysis and Design



PART IV

The Analysis Process #4: Introduction to Functional & Non-Functional Requirements

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■ OBJECTIVES

1. To differentiate users and system requirements
2. To understand functional requirements (FR) and non-functional requirements (NFR)

■ MAJOR TOPICS

REQUIREMENT

- What is a Requirement?
- Types of Requirement

← PART I

FUNCTIONAL REQUIREMENTS

- What is FR

← PART II

NON-FUNCTIONAL REQUIREMENTS

- What is NFR?

← PART III

PART I

- What is a Requirement?
- Types of Requirement

■ What is a Requirement?

It may range from a high-level abstract statement of a **service** or of a system **constraint** to a detailed mathematical functional specification

This is inevitable as requirements may serve a dual function:

- ✓ May be the basis for a bid for a contract - therefore must be open to interpretation
- ✓ May be the basis for the contract itself - therefore must be defined in detail
- ✓ Both these statements may be called requirements

■ Types of Requirement

User requirements

- Statements in **natural language plus diagrams** of the services the system provides and its operational constraints
- Written for **customers**

System requirements

- A **structured document** setting out detailed descriptions of the system's functions, services and operational constraints
- Defines what should be implemented so may be part of a contract between **client and contractor**

■ User And System Requirements

User requirements definition

- 1.** The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

- 1.1** On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
- 1.2** The system shall generate the report for printing after 17.30 on the last working day of the month.
- 1.3** A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
- 1.4** If drugs are available in different dose units (e.g. 10mg, 20mg, etc) separate reports shall be created for each dose unit.
- 1.5** Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

PART II

- Functional Requirements

■ Functional Requirements (FR)

- Describe functionality or system services
- Depend on the type of software, expected users and the type of system where the software is used
- Functional **user requirements** may be high-level statements of what the system should do
- Functional **system requirements** should describe the system services in detail

■ Example: FR in Mentcare System

- A user **shall** be able to search the appointments lists for all clinics
- The system **shall** generate each day, for each clinic, a list of patients who are expected to attend appointments that day
- Each staff member using the system **shall** be uniquely identified by his or her 8-digit employee number

PART III

- Non-functional Requirements

■ Non-functional Requirements (NFR)

- These define **system properties and constraints** e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.
- **Process requirements** may also be specified mandating a particular Integrated Development Environment (IDE), programming language or development method
- Non-functional requirements **may be more critical** than functional requirements. If these are not met, the system may be useless

■ Examples of NFR in the Mentcare System

Product requirement

The Mentcare system shall be available to all clinics during normal working hours (Mon–Fri, 0830–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

Organizational requirement

Users of the Mentcare system shall authenticate themselves using their health authority identity card.

External requirement

The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

■ Metrics for Measuring NFR

Metrics	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

■ SUMMARY

REQUIREMENT

- It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification
- User & System Requirement

FUNCTIONAL REQUIREMENTS

- Describe functionality or system services

NON-FUNCTIONAL REQUIREMENTS

- Define system properties and constraints



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Thank You

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